

EECE 5496 Week 2:

September 10, 2026

1 Instructions for Developing In-Class Model

1. Create a package to organize the model elements
2. Upload the Excel file with the soldier code data: Create element under the package - select Attached File
3. Create BDD with a Block named Battalion that includes by directed composition a Soldier Block which has a multiplicity 1..* . Include in this block two integer valued value properties NsensorA and NsensorB that keep a record of number of soldiers with sensor A on and sensor B on respectively.
4. Create value properties for the Soldier Block that correspond to columns in the Excel file attributing appropriate types (string, integer, real etc.). Also include the two Boolean value properties for Sensor A and Sensor B that may be on or off.
5. Next map the Excel file elements attached in Step 2 to Instances:
 - Select from top menu : 'File' $\rightarrow$ 'Import From'

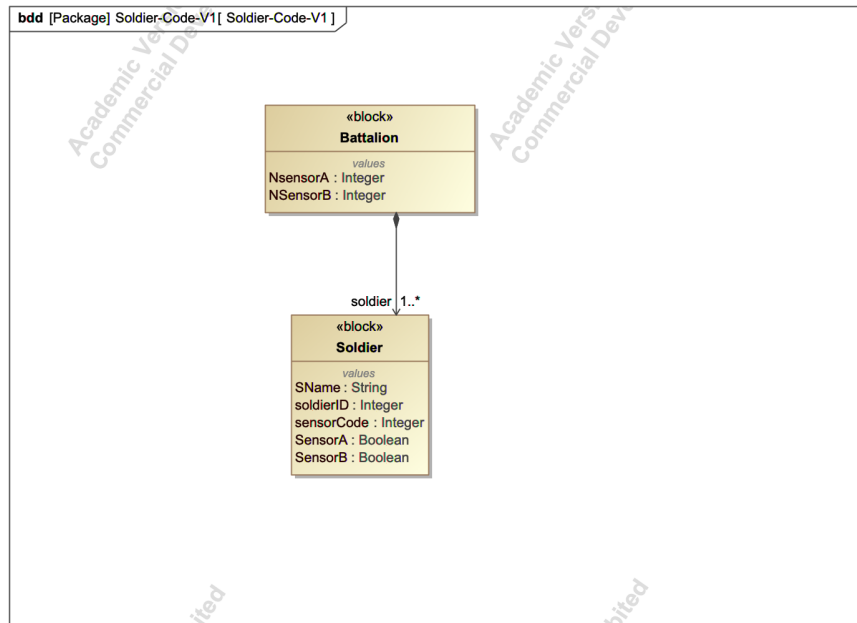


Figure 1: BDD

→ Excel/CSV File → 'Import Using New Map'
which opens up a dialog box.

6. Select the Attached File from the Model.
7. Select 'Import Type' to 'Instance'
8. Select 'Instance Classifier' to Soldier Block
9. Select 'Properties to Map' and select all value properties that are to be aligned with the Excel columns
10. Select 'Target Scope' as the Package where you are storing the instances
11. Connect the Element Properties to the appropriate Excel Columns on the right (drag a connector between the two)

12. Finally click 'Import' : You should see the instances created in the Containment tree
13. First create an Instance for the Battalion Block and select required properties and parts : NsensorA, NsensorB and its Soldier subsystem. Hit 'Finish' to save the instance
14. Selecting the instance just created in the containment tree, select 'Slots' and select the soldier feature, hit the '+' button. From the Instance Specification window that pops up select 'List' - you should see all of the soldier instances from your spreadsheet.
15. Hit the 'multiple selection' icon on the bottom right, select all of the instances you are interested in and hit the '+' sign. They will appear in the right window. Hit 'ok' and close the instance.
16. Next step is to create an Activity Diagram for the Battalion Block that is to iteratively retrieve each Soldier's data and apply it in the context of the problem being modeled. In this problem, we are interested in counting the numbers of sensor A that are on and the numbers of sensor B that are on.
17. In this example, basic Actions some of which may be new to you are introduced. The Activity diagram shown in the figure below includes four types of Action nodes: (i) readSelf; (ii) readStructureFeature; (iii) Expansion Region (shown with the iterative tag); and (iv) Opaque (shown without any tag). These nodes can be accessed from the 'Any Action'

Metaclass.

- ‘Read Self’: To retrieve the current context. If we run the activity diagram with context, ‘self’ will refer to the battalion block.
- ‘Read Structural Feature’: To retrieve the values of a specific structural feature, such as part properties, value properties etc.
- ‘Expansion Region’: To take collection of model elements and perform operations (e.g., actions) on them.
- ‘Opaque’ action: To execute code using languages such as Built-in Math, MATLAB, Groovy etc.

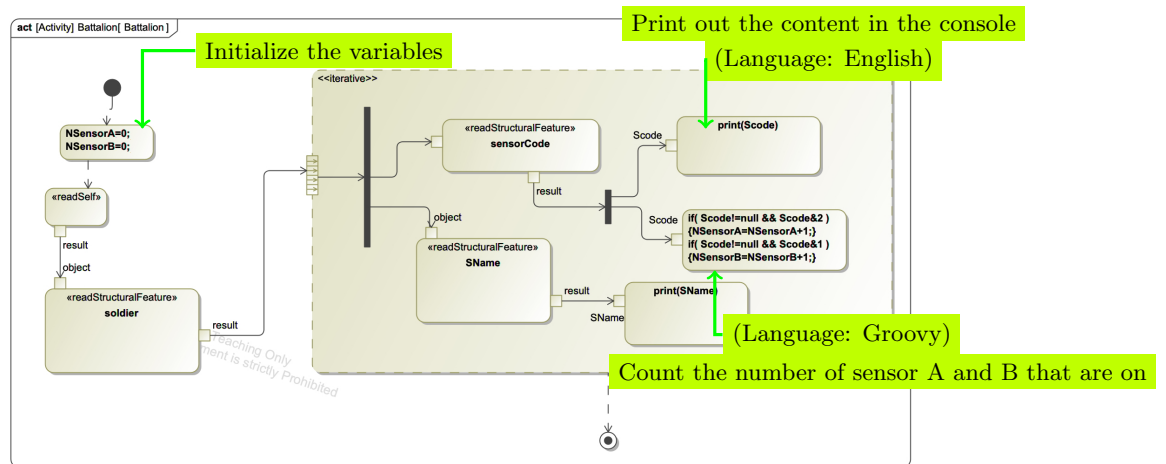


Figure 2: Activity diagram for the Battalion

18. Finally, run the simulation from the Battalion instance and see the code and name values being printed out and the numbers of sensors being counted.